- Design being the core focus of this course we need to understand about its philosophy and area of studies. Design is truly an interdisciplinary fields of study. It produces a new breed of creative people who are keen observer of nature and human behaviour. Designers are social scientists, social anthropologists, scientists, technologists, artists, philosopher, etc. Do you think Concept and Creativity has any link to Design? How and why?

- **What is Design (Industrial Design)?**
- **It is extremely difficult to define 'design' through one single definition. 'Design' word is used in various fields with different meaning and connotation- industrial design, graphic design, carpet design, circuit design, information design, architecture design, social design, business design, pottery design, environmental design, landscape design to software design.**
- Design, itself, is often difficult to define to non-designers because the meaning accepted by the design community is not one made of words. Instead, the definition is created as a result of acquiring a critical framework for the analysis and creation of artifacts. One of the many accepted (but intentionally unspecific) definitions of design originates from Carnegie Mellon's School of Design, "Design is the process of taking something from its existing state and moving it to a preferred state."
- Therefore it is a continuous process evolving new ideas as and when we face problem. A creative act may remain dormant for centuries. It starts changing when a problem is faced and a new creative solution emerges.

What do you do when you face Problem?

Scratch your head! Trying to get idea to solve the problem- pacing up & down- walking, etc.

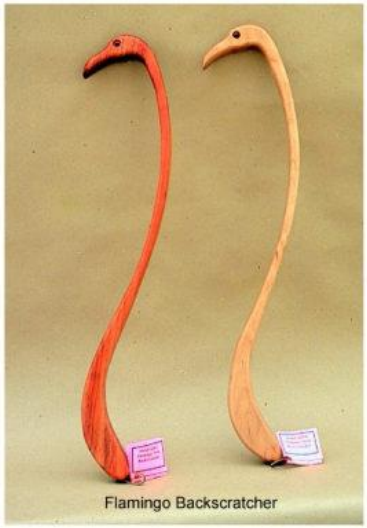
Let us take a very simple-interesting problem- itching. How do you solve the problem?

Concept (solution)

Itching- scratching is the best solution.

However, we are not able to reach every part of the body.
I may have physical problem. In that case what do you do?
Do you try to invent to solve the problem?

The Cat invents its own idea to get relief from itching.

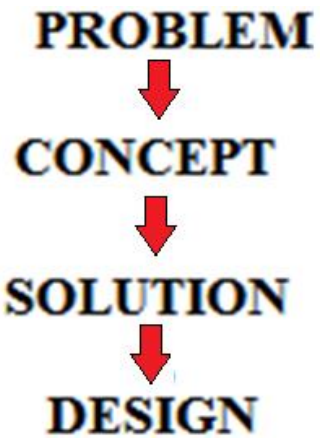


You invent your solution

Scalp scratching device is an excellent innovation. Because it also provides massage to your scalp.

Concept- How do you know scratching would help you? We know by birth that we have a device built in our body- 'nail' that may solve my problem. Past experience helps you to initiate new solutions. Neurological study would prove, how brain sends signal to perform certain way to solve problems.

- Therefore, scratching mechanism helped you to solve a problem. Next question comes- scratching is the only solution for itching or you have multiple solutions? Never think that you have only one solution. As prospective designer, you should have multiple ways to solve a problem. Because human being obvers various alternative conditions to solve a problem.
- You try to solve through some solution. The 'solution' immerges based on a 'concept'. The concept immerges due to earlier experience (scientific principles, other mechanism, function, etc.). The **Concept** creates an idea (s). The idea leads to searching the method to execute/ implement, which produces Design. However, concept may vary from person to person due to cultural exposure.



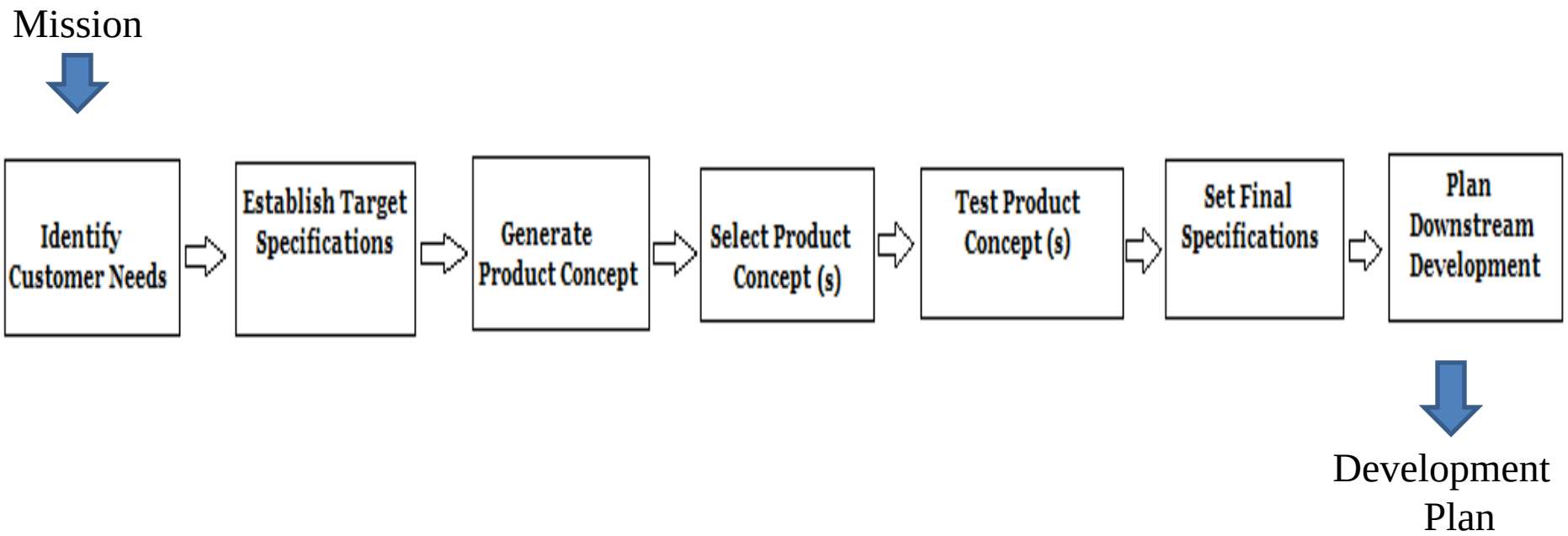
Faced a problem

- Define/ analyze
- Concept is generated
- Solution is found
- Design (Prototype)
- Testing

- Creativity is one of the most important components in design. It follows a trajectory from Problem solving to Concept to Innovative Design.

- **Design students should learn to imagine**. As a child we could all imagine hundreds of things but we do not know when we stopped imagining! Rightfully Einstein had once said that **'…imagination is more important than knowledge**
- **since knowledge has limitation while imagination encircles the world'**. A curriculum or situation should provoke people to imagine, which would motivate to do something differently. Ability to imagine is the key to creative pursuit. Imagination and innovation are two sides of the same coin. Innovation follows imagination.

Problem Identification-Concept Generation- Production



- **<u>Design has to consider the following before launching in the market-</u>**

- **<u>How Important Is Industrial Design to a Product?</u>**
- **All products that are used, operated, or seen by people depends critically on ID for the commercial success.**
- **In order to assess the importance of ID to a particular product is to characterize importance along two dimensions-**
- **1. Ergonomics- encompasses all aspects of product that relates its human interface**
- **<u>Ergonomic Needs-</u>**
- **How important is ease of use?**
- **How important is ease of maintenance?**
- **How many user interactions are required for the product's functions?**
- **How novel are the user interaction needs?**
- **What are the safety issues?**
- **2. <u>Aesthetics Needs-</u>**
- **Is visual product differentiation required?**
- **How important are pride of ownership, image, and fashion?**
- **Will an aesthetic product motivate team?**

Emotional Appeal-

- **- Is the product attractive? Is it exciting?**
- **- Does the product express quality?**
- **- What images come to mind when viewing it?**
- **- Does the product inspire pride of ownership?**
- **- Does the product evoke feelings of pride among the development team and sales staff?**

- **<u>Examples of product-specific questions include</u>:**
- **- How does the car door sound when slammed?**
- **- Does the hand-tool feel solid and sturdy?**
- **- Does the appliance look good on the kitchen counter?**

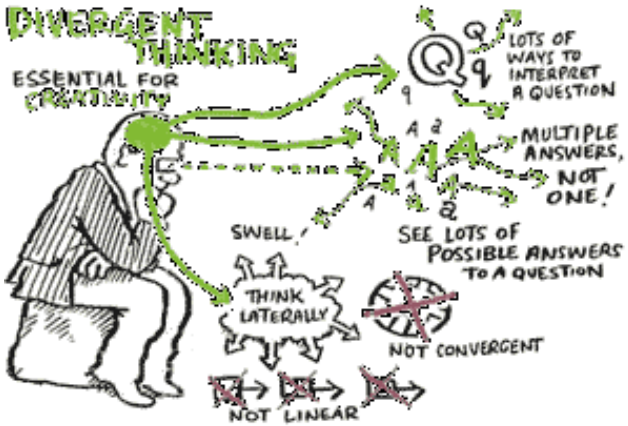
- **<u>Product Differentiation</u>**
- **- Will a customer who see the product in a store be able to identify it because of its appearance?**
- **- Will it be remembered by a consumer who has seen it in advertisement?**
- **- Will it be recognized when seen on the street?**
- **- Does the product fir with or enhance the corporate identity?**

Human being couldn't squeeze orange by hand efficiently. It requires strength that many people may not have. Can you think of a device which will provide easier solution? Think of a product design, using the same principle of squeezing? Each problem gives opportunity for a new innovation.

"A leading business school did a study that showed that its graduates performed well at first, but in ten years, they were overtaken by a more streetwise, pragmatic group. The reason according to the professor who ran the study: 'We taught them how to solve problems, not recognize opportunities. When opportunities knocked, they put out their 'Do not Disturb' signs." (Roger Von Oech, **A Whack on the SIDE of the Head**, London: Harper Collins Pub. 1990, p.27)

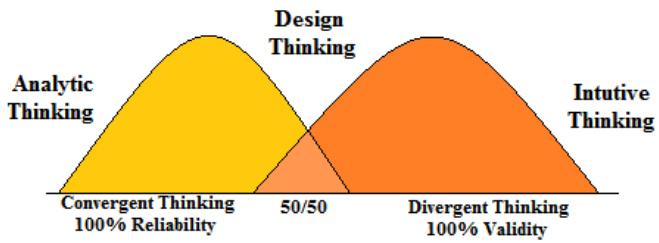
- Thus, able to imagine is the key to design profession. Obvious answers limits or restrict imagination to run free. Multiple choices or probable solutions widen the scope of unique solution. Creative thinking requires an attitude that allows one to search for ideas and manipulate your knowledge and experience. Taking unusual steps opens up new avenues or possibilities.

- Albert Szent-Györgui the Nobel Prize winning physician said, **“Discovery consists of looking at the same thing as everyone else and thinking something differently.”**
- Looking differently is a typical quality of a designer. ‘Inquisitiveness’ is one very important quality of design profession. Some time it may be child like innocence and curiosity that inspires you to be creative designer.
- Let us try to understand the process of ‘creative thinking’. There are two types of ‘Creative Thinking’ –

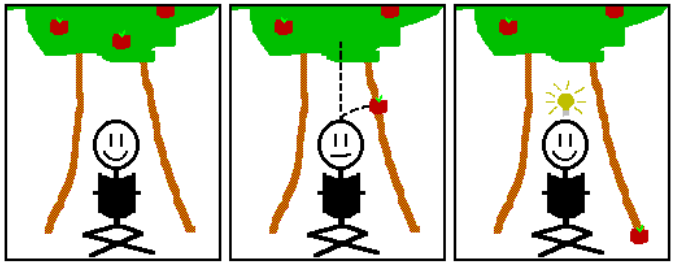


- **Intuition**
- **Subjective**
- **Emotional**
- **Imaginative**
- **Impulsive**
- **Holistic**
- **Free-wheeling**
- **Qualitative**

- ***Divergent Thinking*** is the intellectual ability to think of many original, diverse, and elaborate ideas. It is a thought process or method used to generate creative ideas by exploring many possible solutions.
- ‘Divergent Thinking’ helps to thinking multiple ways to solve a problem. Therefore, always try to come up with multiple options to solve a problem. No option should be connected to the other one. Each one should have unique characteristic. Therefore, consciously avoid linear solution which creates limitation. Divergent Thinking allows you to be ‘dreamer’ (remember what Einstein said?) with new ideas. As a designer your thinking pattern should be quite different than others.



Logical
Objective
Intellectual
Realistic
Planner
Discriminative
Structured
Quantitative

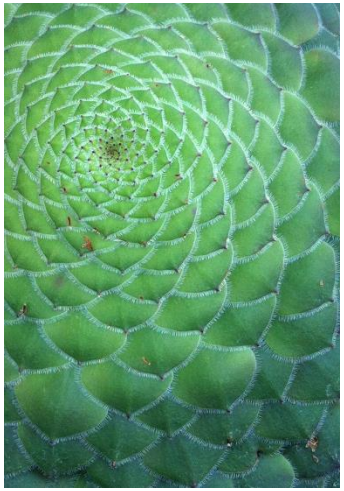


Law of Motion

- ***Convergent thinking***: the intellectual ability to logically evaluate, critique and choose the best idea from a selection of ideas.
- "Convergent thinking is the type of thinking that focuses on coming up with the single, well-established solution to a problem. It is oriented toward deriving the single best, or most often conclusive answer to a question. Convergent thinking emphasizes speed, accuracy, and logic and focuses on recognizing the familiar, reapplying techniques, and accumulating stored information. It is most effective in situations where an answer readily exists and simply needs to be either recalled or worked out through decision making strategies. A critical aspect of convergent thinking is that it leads to a single best answer, leaving **no room for ambiguity**. In this view, answers are either right or wrong. The solution that is derived at the end of the convergent thinking process is the best possible solution the majority of the time."
- (Source: https://en.wikipedia.org/wiki/Convergent_thinking Jan. 13, 2016)

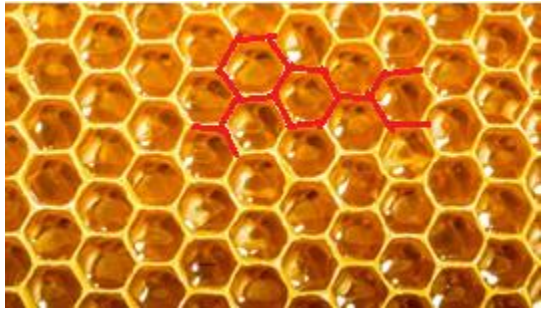
- **Characteristics of the creative personality:**
- **Creative individuals have a great deal of energy, but they are also often quiet and at rest**
- **Creative individuals tend to be smart, yet also naive at the same time**
- **Creative individuals have a combination of playfulness and discipline, or responsibility and irresponsibility**
- **Creative individuals alternate between imagination and fantasy at one end, and rooted sense of reality at the other**
- **Creative people seem to harbor opposite tendencies on floating between extroversion and introversion**
- **Creative individuals are also remarkable humble and proud at the same time**
- **Creative individuals to a certain extent escape rigid gender role stereotyping and have a tendency toward androgyny (favourable towards both sex)**
- **Generally, creative people are thought to be rebellious and independent**
- **Most creative persons are very passionate about their work, yet they can be extremely objective about it as well**
- **The openness and sensitivity of creative individuals often exposes them to suffering pain yet also a great deal of enjoyment**
- **Creative people never take extreme stand**

- **Where do we look for innovative ideas and resources?**



- Mother ‘nature’ is the greatest source of innovative ideas that human being can search for. It is an amazing experience when you observe the natural phenomenon. ‘Biomimicry’ is the field of study that focuses on nature and its behaviour. Human being have learned to mimic nature’s behaviour apply it in design.

- Following are the basic observations-
- **Basic Form**
- **Functional Relationship** (how two parts relate to each other)
- **Behaviour** (how each part behaves in respect to other part)



Beehive Structure

Modular Structure